Li Wang 王力

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Homepage · Google Scholar



Research Interests: Audio safety, audio deepfake detection, and audio language models

Education

The Chinese University of Hong Kong, Shenzhen — Ph.D. in Computer Science Sep. 2023 – Jul. 2027 (Expected)

Audio deepfake detection and anti-spoofing; GPA 3.73/4.00, top 5%; Advisors: Zhizheng Wu and Haizhou Li

Peking University — M.Eng. in Computer Application Technology (Exam-exempted) Sep. 2019 – Jul. 2022

Keyword spotting and speaker verification; top 5%; Advisor: Yuexian Zou

University of Electronic Science and Technology of China — B.Eng. in Software Engineering Sep. 2015 – Jun. 2019

GPA 3.69/4.00, ranked 8/162 (top 4.9%)

Selected Publications

1. **Li Wang** et al. *GenTraceBench: A Benchmark for Tracing Audio Deepfakes Across Pre- and Post-training Stages.* **ISCSLP 2026, accepted.**
2. **Li Wang***, Jindong Wang*, Wan Lin*, Kunyu Feng* et al. *RealComm: Benchmarking and Adapting Audio Deepfake Detection over Real Communication Channels.* Submitted to **IEEE/ACM TASLP**. (* Equal contribution.)
3. Yupei Li*, **Li Wang*** et al. *DFALLM: Achieving Generalizable Multitask Deepfake Detection by Optimizing Audio LLM Components.* Technical report, 2025. (* Equal contribution.)
4. Xueyao Zhang et al. *SpeechJudge: Towards Human-Level Judgment for Speech Naturalness.* **ICLR 2026**.
5. **Li Wang** et al. *Over-the-Air Adversarial Attacks and Detection for Automatic Speaker Verification.* **IEEE/ACM TASLP, 2026**.
6. Peizhuo Liu*, **Li Wang***, Renqiang He* et al. *SpMis: An Investigation of Synthetic Spoken Misinformation Detection.* **SLT 2024, Best Paper Finalist (top 2.5%).**
7. **Li Wang** et al. *AdvSV: An Over-the-Air Adversarial Attack Dataset for Speaker Verification.* **ICASSP 2024**.
8. **Li Wang** et al. *Learning Decoupling Features Through Orthogonality Regularization.* **ICASSP 2022**.
9. **Li Wang** et al. *Text Anchor Based Metric Learning for Small-Footprint Keyword Spotting.* **Interspeech 2021**.

Additional work: Teffic-Audio, VoxSafeBench, and Neural Replay Simulator. Full list: Homepage / Google Scholar.

Professional Experience

Shenzhen Research Institute of Big Data (SRIBD) — Research Assistant Nov. 2022 – Aug. 2023

Led a Huawei collaboration on physical attacks against speaker verification; built a 1.1M-sample attack dataset and established the defense approach adopted for continued development.

International Digital Economy Academy (IDEA) — Financial ML Engineer Jul. 2022 – Oct. 2022

Developed financial factors and deep-learning models for stock-return prediction.

Internships

ByteDance AI Lab — Research Intern Jun. 2021 – Mar. 2022

- Worked on audio fingerprinting and music segmentation; proposed a temporal localization algorithm that improved accuracy from 95.80% to 97.90% and achieved 0.62 temporal IoU.

Huawei MTI, Audio Engineering Department — Research Intern Apr. 2021 – Jun. 2021

- Reproduced and evaluated D3Net on MUSDB for vocal, drum, bass, and residual-stem separation, achieving 6.43 average SDR.

Tencent AI Lab — Research Intern May 2020 – Nov. 2020

- Evaluated GALR speech separation for downstream speaker verification and curated speaker metadata for 2M audio samples.

Selected Projects

Huawei HarmonyOS 7 Audio Deepfake Detection — Contributor Jan. 2025 – Jan. 2027 (Planned)

- Contributed to the research, development, and product deployment of the audio deepfake detection capability released in HarmonyOS 7.
- Worked on Chinese deepfake data construction, telephone-channel robustness, lightweight-model evaluation, and delivery of an executable inference package.

AI Voice Wake-up Technology with Xiaomi — Project Lead Jan. 2021 – Jan. 2022

- Led industry communication, experimental planning, and model/data-pipeline optimization for wake-word and personalized wake-up tasks.
- Reduced false rejection from 20% to 2% and achieved 99.7% accuracy; the work supported a first-author ICASSP 2022 paper.

Patents

First inventor (co-inventor: Zhizheng Wu; assignee: SRIBD):

1. Audio deepfake detection method, apparatus, computer device, and storage medium. **ZL202411863864.6, granted Nov. 2025**.
2. Method and system for detecting adversarial audio with pretrained models. **Application No. 202410942156.5**, pending.
3. Open-set audio deepfake algorithm attribution method. Application No. **202510885063.8**, pending.

Academic Service

Reviewer: IEEE/ACM TASLP, ICASSP, IEEE SPL, Computer Speech & Language, ASRU, IALP, and NCMMSC.
Standards: Contributor to TC260's method for identifying AI-generated synthetic content (implicit metadata identification for text files).
Teaching Assistant: Programming Methodology; Speech and Language Processing; Machine Learning (2023–2025), CUHK-Shenzhen.
Student Volunteer: IEEE Spoken Language Technology Workshop (SLT 2024).

Honors and Skills

Honors: SLT 2024 Best Paper Finalist (top 2.5%); Best Poster, Speaker Recognition Research and Application Workshop (2024); Duan Yongping Travel Scholarship (2023); National Second Prize, National University Student Information Security Competition (2018); First-Class Undergraduate Scholarship (2015–2017); Outstanding Volunteer, 9th National Games for Persons with Disabilities (2015).
Skills: Python, PyTorch, Java, C++; native Mandarin, technical English; Erhu Grade 10.